



Pipe dreams: Cannabis legalization and the persistence of racial disparities in jail incarceration

Guangzhen Wu^{*}, Katherine A. Durante, Heather C. Melton

Department of Sociology, The University of Utah, 380 S 1530 E Rm301, Salt Lake City, UT 84112-0250, United States of America

ARTICLE INFO

Keywords:

Recreational cannabis legalization
Jail incarceration
Racial disparities
Racialized social control
Synthetic control method

ABSTRACT

There is an emerging body of research examining the impact of adult-use recreational cannabis legalization on racial disparities in arrests, yet this is among the first studies to examine changes to the jail population composition. We use a quasi-experimental design, including difference-in-differences (DiD) analysis and matching via the synthetic control method (SCM) to examine the impact of recreational cannabis legalization on Black–White disparity in jail populations in Colorado and Washington State. Our findings reveal that Black–White disparity has overall declined in both legalized and criminalized states. We observe more of a reduction in Black–White disparity in legalized states; however, the results are not substantially or statistically significantly different (DiD results: Estimate = -0.052 , $SE = 0.320$, $p = 0.872$; SCM results being consistent with the DiD results). This research provides important insight into continued racialized social control even when laws become liberalized. Further, the findings of this study have important policy implications, as voters and legislative bodies should seek policies more purposefully designed to reduce racial disparity in the jail population if this is their goal.

1. Introduction

About 30% (514,000) of people incarcerated in the United States on any given day are in local jails, and 1 in 5 are there for drug offenses (Sawyer and Wagner, 2023). Cannabis-related crimes continue to be the most common drug offenses that lead to arrests (American Civil Liberties Union [ACLU], 2020); there were nearly 700,000 cannabis arrests in the United States in 2018 alone, which accounted for over 40% of all drug arrests and surpassed the total number of arrests for all violent crimes combined (ACLU, 2020). Black Americans are about 3.64 times more likely to be arrested for cannabis possession even though they consume it at similar rates as their White counterparts (ACLU, 2020). Consequently, they disproportionately experience the many adverse penalties—such as legal fees, time in court and carceral facilities, stigma, and reduced opportunities for employment, housing, and education (Alexander, 2010)—related to criminal legal system involvement for drug offenses. Cannabis legalization, then, presents an opportunity to address historic racial harm, especially if policy changes include retroactive plans to expunge records, reduce current sentences, and center racial justice in the burgeoning cannabis industry (Cooper and Thompson, 2020).

However, even as overall cannabis arrests have plummeted in jurisdictions that have legalized it, racial disparity persists, and in some jurisdictions, the racial arrest disparity for cannabis has even increased (Firth, Maher, Dilley, Darnell, and Lovrich, 2019; Joshi, Doonan, and Pamplin II, 2023). For example, Joshi et al. (2023) found that in Los Angeles, California, the Black–White disparity in total cannabis arrests increased to a ratio of 12.7 compared to the pre-legalization disparity ratio of 7.6, despite total cannabis arrests dropping by >94%. Still, the effects of legalization remain under-examined, despite the policy-relevant implications; the impact of recreational cannabis legalization on crime, arrests, and incarceration—including jail incarceration—and racially disparate outcomes remains unclear.

Although the limited existing research generally shows no significant reductions in racial disparities in cannabis/drug arrests after legalization, it does not necessarily mean that there will be no meaningful changes to the racial disparity in jail populations. A dramatic reduction in the volume of cannabis /drug arrests may result in a substantial reduction in racial disparity in overall jail incarceration even if disparity persists. Further, the lenient approach used in cannabis law enforcement in the context of legalization may “spill over” in some way to the policing of other illicit drugs, leading to declining trends of drug arrests

^{*} Corresponding author.

E-mail addresses: guangzhen.wu@soc.utah.edu (G. Wu), katherine.durante@soc.utah.edu (K.A. Durante), heather.melton@soc.utah.edu (H.C. Melton).

<https://doi.org/10.1016/j.jcrimjus.2024.102230>

Received 31 May 2024; Received in revised form 15 July 2024; Accepted 15 July 2024

Available online 24 July 2024

0047-2352/© 2024 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

(Wu and Cullenbine, 2022). Alternatively, law enforcement may adapt to increase arrests for other offenses once many cannabis-related behaviors are no longer arrestable (Makin et al., 2019). Hence, it is necessary to examine if cannabis legalization has affected the racial disparity in incarcerated jail populations. This exploration is policy-relevant and can inform discussions of the complex dynamics of racialized social control, and offer empirical evidence for state legislatures and voters considering cannabis legalization to assess if it has served to reduce racial disparity in carceral facilities.

In this study, we employ a quasi-experimental design involving the synthetic control method (SCM) and the difference-in-differences (DiD) analysis to examine the Black–White incarceration disparity in large jails in Colorado (CO) and Washington State (WA) both before and after the legalization of adult-use recreational cannabis, in comparison to states that have not legalized it. Notably, a lack of administrative data on incarcerated populations that consistently classify information across jurisdictions for Hispanics/Latinos and non-Hispanics/non-Latinos and groups such as Asians and Native Americans precludes us from analyzing other racial and ethnic groups (Durante, Trocchio, and Martínez, 2024). CO and WA were chosen because they were the first to legalize recreational use in 2012, allowing for an extended observation period to assess potential changes in the outcome variable.

2. Literature review

2.1. Background

The criminalization of cannabis in the United States is rooted in the racist early 20th-century misperception that people of color, especially Black and Mexican men, who consume it have a high propensity for committing violent acts (Bender, 2016). Later, the Nixon administration, upon declaring a “war on drugs,” sought to culturally tie Black Americans and antiwar liberals to cannabis and other illicit substances to vilify them as a means of political assault (LoBianco, 2016). Viewed from this lens, scholars have argued that more recent drug control policies, including harsh criminal penalties adopted in the 1980s and 1990s, serve as a political mechanism for targeting racially minoritized individuals, particularly Black men, and creating an underclass (Alexander, 2010; Tonry, 1995). Thus, the continued criminalization of cannabis today can be understood as a means of surveillance and social control of Black and Brown communities, as members of these communities are unfairly targeted and subjected to stops, frisks, arrests, and convictions (ACLU, 2020; Alexander, 2010). However, cannabis laws are rapidly changing across the United States, and recreational legalization has the potential to reduce police contacts, arrests, exposure to carceral facilities, and the development of a criminal history for Black Americans, especially.

CO and WA became the first states to legalize adult-use recreational cannabis in late 2012. To date, 24 states and the District of Columbia (DC) have adopted laws to legalize recreational cannabis possession, production, and retail sales. About 60% of U.S. residents report that adult-use recreational cannabis should be legal, with support being over 70% among adults under age 30 (Green, 2022). Growing public support paired with the financial benefits through the generation of tax dollars (Dills, Goffard, Miron, and Partin, 2021) and efforts at the federal level to reclassify cannabis as a less dangerous drug (Lieb, 2024) suggests that more states are poised to follow this trend. Still, it should be acknowledged that resistance to legalization remains; only 46% of Americans over age 65 and 45% of Republicans support legalization (Green, 2022). Further, fewer than 20% of Republican voters view “reducing bias against minorities” as a policy priority for the U.S. criminal legal system (Brenan, 2023). This opposition may hamstring any efforts to legalize cannabis and craft policies aimed at reducing racial disparities.

2.2. Cannabis legalization and crime

People in jail are typically accused of committing one or more crimes. Recreational cannabis legalization is theorized to have both direct and indirect effects on criminal activity. First, legalization allows people to purchase and consume cannabis without the fear of criminal sanctions, thereby, directly reducing criminal activity by redefining these behaviors as legal. Next, the increase in cannabis availability and the lack of legal consequences may make it an attractive alternative to alcohol, which may indirectly reduce crime and violence associated with alcohol consumption (Brinkman and Mok-Lamme, 2019; Dragone, Prarolo, Vanin, and Zanella, 2019; Zellers et al., 2023). Further, the underground drug market may be diminished as a result of legalizing a formerly illicit substance, both directly and indirectly reducing crime (Brinkman and Mok-Lamme, 2019; Dragone et al., 2019). Finally, legalization may indirectly reduce crime by improving community safety by changing police behavior; legalization should facilitate law enforcement to allocate their time and resources to more serious offenses (Dragone et al., 2019; Makin et al., 2019; Wu, Li, and Lang, 2022).

Several studies offer support for the crime reduction hypothesis resulting from legalization. For example, recreational cannabis legalization is associated with increased cannabis use and a decrease in alcohol use (Dragone et al., 2019; Zellers et al., 2023) as well as crimes often associated with alcohol use, including a reduction in rape (Dragone et al., 2019). Legalization has also been found in some studies to improve crime clearance rates, as police focus their time and resources on more serious offenses (Makin et al., 2019; Wu et al., 2022), though scholars also observed inconsistent findings (Jorgensen and Harper, 2020). At the neighborhood level, research has been mixed as to whether cannabis dispensaries are associated with any change in criminal activity (Brinkman and Mok-Lamme, 2019; Contreras, 2017; Freisthler, Gaidus, Tam, Ponicki, and Gruenewald, 2017). Similarly, no clear pattern has emerged regarding changes to index crimes in jurisdictions that have legalized recreational-use cannabis (Harper and Jorgensen, 2023; Lu et al., 2021; Maier, Mannes, and Koppenhofer, 2017; Wu, Wen, and Wilson, 2021; Wu and Cullenbine, 2022; Wu and Willits, 2022), nor in their adjacent jurisdictions where it remains illegal (Brinkman and Mok-Lamme, 2019; Wu, Boateng, and Lang, 2020). Overall, research has yielded mixed and inconclusive results regarding the effects of cannabis legalization on crime. Thus, it remains unclear how cannabis legalization may affect jail populations, given that criminal involvement is a key mechanism leading to an arrest and subsequent jail admission.

2.3. Cannabis legalization and arrests

Arrests for cannabis-related offenses have, as expected, plummeted in jurisdictions that have adopted laws that legalize or decriminalize adult-use cannabis (Farley and Orchowsky, 2019; Firth et al., 2019; Hao and Cowan, 2020; Joshi et al., 2023; Meize et al., 2022; Sheehan, Grucza, and Plunk, 2021; Tran et al., 2020; Willits et al., 2022; Wu and Cullenbine, 2022; Wu and Willits, 2022). The total decline in arrests holds across members of different racial groups, and yet, unfortunately, racial disparity in cannabis arrests persists (Farley and Orchowsky, 2019; Firth et al., 2019; Hao and Cowan, 2020; Meize et al., 2022; Sheehan et al., 2021; Tran et al., 2020; Willits et al., 2022), and in some cases, has increased (Firth et al., 2019; Joshi et al., 2023). Notably, research on the spillover effects of legalization on cannabis-related arrests does not find such a distinct pattern; no discernable trend in changes to arrests has been found in studies of neighboring jurisdictions where cannabis remains illegal (Ellison and Spohn, 2017; Farley and Orchowsky, 2019; Hao and Cowan, 2020; Wu et al., 2020). The mass reduction in cannabis arrests paired with continued racial disparity therein underlies the importance of examining the impacts of cannabis legalization on jail incarcerations.

2.4. Cannabis legalization and jail bookings

Changes in patterns of arrests should directly impact patterns of jail admissions, and we would expect then that the overall number of people admitted to jails would decline as cannabis arrests decline. Still, it is unclear whether this would translate into a reduction in the racial disparity for total jail incarcerations, and there is a dearth of research on the effects of legalization on jail use. For example, despite disproportionate arrests, the racial disparity in jail populations might still decline because cannabis offenses might lead to fewer jail bookings (Stanton et al., 2020) as cannabis becomes more socially acceptable in both legal and non-legal states alike. Further, the reallocation of police resources may lead to increased arrests and jail bookings for other, non-cannabis-related offenses (Makin et al., 2019; Wu et al., 2022). In an analysis of WA, Stanton et al. (2020) found that legalization has not had a substantial or consistent impact on jail bookings.¹ Similarly, in a recent study based on the National Corrections Reporting Program (NCRP) data, Wu et al. (2023) found no significant effects of recreational cannabis legalization on Black–White disparities in overall prison incarceration or drug-related prison incarceration. Notably, that study only focused on prison populations; cannabis policy reform should be more impactful for jail incarceration given that legalization has a more direct impact on low-level offenses (Wu and Willits, 2022). Put differently, if legalization were to have an effect, we would expect to be able to see it more clearly in jail populations.

The absence of a reduction in jail admissions in WA (Stanton et al., 2020) raises questions about legalization as a means for reducing disparity, especially as racial disparity in arrests for cannabis-related offenses persists. Still, the total number of arrests for cannabis-related offenses has declined, which should directly impact the jail population size and demographics. Indeed, several competing rationales suggest that the racial disparity in jail admissions should be impacted by adult-use recreational cannabis laws.

2.5. Theorizing continued racial disparity

Legalizing cannabis may be particularly impactful on policing behavior— and therefore, arrests and jail incarcerations— since the prohibition of cannabis helped to make pretextual stops commonplace among law enforcement (Kreit, 2016). Indeed, cannabis use is so common in U.S. society that it is plausible that the substance or related contraband will be found when law enforcement searches vehicles or pedestrians on a hunch. Critics of pretextual stops argue that the criminalization of cannabis 1) grants law enforcement nearly unlimited power to pull over drivers, 2) infringes on individual liberties, and 3) causes racial disparities in who is pulled over, searched, arrested, and admitted to jail (Kreit, 2016). Limits to pretextual stops should force police to focus on other offenses, therefore impacting who is arrested and admitted to jails.

Racial disparity in drug arrests is largely based on police discretion and decision-making, and therefore, the racial disparity in jail populations may persist even post-legalization. Several cannabis-related offenses remain illegal in states that have legalized recreational use,

such as driving while under the influence, possession under the age of 21, and public consumption (Bender, 2016). Further, high taxation has caused some consumers to continue to illegally purchase cannabis on the underground market (Bender, 2016), thus potentially leading to continued racial disproportionality in arrests and jail incarcerations. For example, an analysis of police stops found that legalization reduced the number of vehicle searches for White, Black, and Latino individuals, but that the bar for searching the latter two groups remained lower (Pierson et al., 2020). Racial bias or discrimination targeted at the Black population may also impact who is arrested for similar offenses or even for the same offense (Lantz, Wenger, and Craig, 2023). Indeed, given the racialized history of the war on drugs, liberalized cannabis policy may just lead law enforcement and state policy to shift focus toward alternative means that uphold racial hierarchy at the expense of Black Americans (Alexander, 2010) especially as little is done to address mass poverty or the collateral consequences of a criminal record, and drug crimes are largely associated with people of color (Beckett, Nyrop, Pfingst, and Bowen, 2005).

In sum, previous theorizing and research suggest that cannabis legalization should change police behavior, therefore opening up the opportunity for reduced disparities in arrests and jail incarceration. However, given that drug enforcement is largely racialized and discretionary, it is also possible that shifting law enforcement attention to other offenses will keep racially disparate outcomes intact. However, as cannabis becomes more socially acceptable, fewer people may be admitted to jail after arrest for related offenses. Further, it remains unclear how law enforcement officials shifting focus to other offenses will impact the racial composition of jail admissions. This study provides important insight into whether adult-use recreational cannabis legalization has significantly or substantially reduced the racial disparity in jail admissions in CO and WA.

3. Methodology

3.1. Data

We use jail population data to investigate the potential effects of recreational cannabis legalization on Black–White jail incarceration disparity in CO and WA through the comparison of trends in the Black-to-White ratio of jail incarceration rates (Black jail rate divided by the White jail rate) for large county jails between the two legalized states and those (34 states) that have not legalized cannabis for recreational purposes (by December 2019).² Specifically, the jail confinement data for this study were based on the Bureau of Justice Statistics (BJS)'s Census of Jails (COJ) for the years 2013 and 2019, and the Annual Survey of Jails (ASJ) for the years 2007–12 and 2014–18 (thus, the study period covers 2007–2019), which we obtained from the Institute for Social Research at the University of Michigan (ICPSR).

The COJ is conducted every five to eight years and covers all jails in the United States and provides the sampling frame for the ASJ. The ASJ uses a stratified probability sampling approach based on jail population data collected through the most recent COJ (in 2018 the ASJ sampled 882 local jails, accounting for approximately one-third of all jails in the

¹ It should be noted that although this study took an important step in advancing our understanding of the effect of legalization on racial disparities in jail populations, it is limited in that it only examined WA, with limited number of jails (8 jails examined, with jail population data missing for the jail in the most populous county – King county) and a relative short observation period (2009–2016) considered. In addition, this study did not use a comparison group (e.g., jails in the non-legalized states). The current study extends the scope to include both WA and CO, with an extended observation period from 2007 to 2019. It is also based on a more rigorous quasi-experimental research design that involves the use of a control group (jails in a number of non-legalized states) to help strengthen causal inference.

² These 34 states are: Alabama, Arkansas, Arizona, Florida, Georgia, Idaho, Illinois, Indiana, Iowa, Kansas, Kentucky, Louisiana, Maryland, Minnesota, Mississippi, Missouri, Nebraska, New Hampshire, New Jersey, New Mexico, New York, North Carolina, Ohio, Oklahoma, Pennsylvania, South Carolina, South Dakota, Tennessee, Texas, Utah, Virginia, West Virginia, Wisconsin, and Wyoming. It should be noted that Connecticut, Delaware, Hawaii and Rhode Island are also eligible to be included as control states. However, these four states have combined prison-jail systems (meaning their incarceration data do not differentiate jail inmates from prison inmates), and their state-operated facilities were excluded from COJ and ASJ. Hence, they were not included in the analysis of the current study.

country) and is conducted during the years between census collections (i.e., ASJ data is unavailable for the years during which COJ was conducted). In selecting jails, the ASJ groups all jurisdictions into 10 strata based on their inmate population size and the presence of juveniles, with strata 1,³ 2, and 7 categorized as large jails with an average daily population (ADP) of over 200. These are also the jails with the highest probabilities of being surveyed across all the waves of ASJ examined in this study. The COJ and ASJ data for the period of 2007–2019 were compiled, and large jails with the stratum of 1, 2, or 7 that appeared in all years of the ASJ and COJ data during the observation period were identified and included in the final dataset⁴ to model the relative long-term trend of racial disparities in jail populations and ensure consistency in the comparisons of racial disparity trends between the treatment and control groups. Limiting jails included in the study to those surveyed across all the years of the observation period is a common approach when using administrative data on crime and incarceration collected by federal agencies for related trend analysis (see Lu et al., 2021; Makin et al., 2019; Willits et al., 2022; Wu and Cullenbine, 2022) because it avoids biasing the results with the inclusion of units that enter and leave the reporting pool during the period of observation (Willits et al., 2022). Jail incarceration data on differential racial groups were aggregated at the county level⁵ for counties with more than one facility. The cleaned jail population dataset included 156 jail jurisdictions in the control groups, 5 jail jurisdictions in CO, and 9 jail jurisdictions in WA. It should be noted here that the 9 jail jurisdictions in WA included King County (i.e., Seattle), though data for this county missed the years 2013, 2015, and 2019. Doing so was to ensure that this most populous county in WA was considered in the analysis.⁶

The cleaned jail population dataset was then merged with county resident population data. Specifically, county-level annual population estimates by race and ethnicity data from 2007 to 2019 were obtained from the U.S. Census Bureau's website. This combined dataset allowed for computing the Black jail rate and White jail rate (dividing the total number of Black/White inmates by the Black/White population in the jurisdiction), which the key measure of this study—Black–White jail incarceration disparity—was created (reflected by the ratio of the Black jail incarceration rate to their White equivalent).

Prior research has demonstrated the relevance of social and

economic conditions to the criminal legal process and the composition of the incarcerated population (Durante, 2020; Western, 2006). Hence, this study also took into account the potential influence of county-level characteristics including the poverty level, median household income, unemployment rate, and percentage of men. Yearly data for these variables, except the percentage of men (information on this came from the aforementioned county resident population dataset), were acquired from the U.S. Census Bureau's Small Area Income and Poverty Estimates (for county poverty level and median household income) and the Bureau of Labor Statistics (under the Department of Labor's Local Area Unemployment Statistics (for county unemployment rate).

Table 1 presents the summary statistics for jail jurisdictions in CO, WA, and those in the nonlegalized states, respectively. Given that both CO and WA passed their recreational cannabis laws in late 2012, the pre-legalization period ends in 2012, and the post-legalization period begins in 2013 (for WA, the post-legalization period begins in 2014, due to the missing data for 2013). As Table 1 displays, Black–White jail incarceration disparities generally decreased in the jail jurisdictions of both the two legalized states and those in the control states, reflected by the lower Black–White jail ratios for the post-legalization periods than those for the pre-legalization periods. However, the decreases in racial disparities appear larger for the two legalized states than that for the control states. For the comparison between CO and the control states, while CO experienced a decrease from the Black-to-White ratio of 4.685 to 3.640 (difference: 1.045), the control states saw a decline from 6.128 to 5.405 (difference: 0.723). Similarly, while WA's Black-to-White jail incarceration ratio dropped from 5.608 to 4.457 (difference: 1.151), the ratio for the control states declined by 0.781 (pre-legalization, 6.128; post-legalization, 5.347). Despite the pattern showing relatively larger decreases in Black–White disparities in large jails in the two legalized states, it is unclear if these decreases are substantial, or happened by chance, which should be addressed by subsequent more advanced analytical models for comparative case studies (e.g., the synthetic control method).

3.2. Analytical strategy

We employed the difference-in-differences (DiD) method to explore the potential link between legalization and Black–White disparity in jail incarceration. In the DiD models, a jail jurisdiction variable and a time variable (year) were added to fit the two-way fixed-effect model.

Incorporating two-way fixed effects in the DiD analyses allows for unobservable heterogeneity in both dimensions of the data: cross-section and time. The DiD method relies upon observations collected from two groups (treatment group and control group) differentiated by the exposure factor (i.e., legalization) within two time periods (pre-treatment period and post-treatment period). The treatment group is the one subjected to the treatment in the second period (post-treatment period), while the control group is not exposed to the treatment during either period. The treatment effect is estimated by extracting the average gain over time in the control group from the gain over time in the treatment group as the same units within a group are observed in each period. The DiD model was specified as follows:

$$Y_{jt} = \alpha + \beta DT_{jt} + \gamma DP_{jt} + \delta (DT_{jt} \times DP_{jt}) + X_{ct} + \theta_j + \tau_t + \varepsilon_{jt} \quad (1)$$

where Y_{jt} (the outcome variable) represents Black–White jail incarceration disparity (the ratio of Black jail incarceration rate to their White equivalent) in jail jurisdiction j in year t . α is the constant. DT_{jt} is an indicator variable for the group-specific component, equal to 1 if Y_{jt} is an observation from the treatment group (jail jurisdictions in CO and WA) and equal to 0 otherwise. DP_{jt} is the dummy variable for the time-specific component, coded as 1 if the observation is from the post-treatment (legalization) period or 0 otherwise. Our key independent variable is $DT_{jt} \times DP_{jt}$, an interaction term indicating the policy impact on Black–White jail incarceration disparity. X_{ct} is a vector of county-level

³ Stratum 1 is referred to as certainty stratum, consisting all jail jurisdictions (257 in 2018) that held juvenile inmates at the time of the 2013 COJ and had an ADP of 500 or more inmates during the 12 months ending on December 31, 2013, and that held adult inmates and had an ADP of 750 or more. All jurisdictions in stratum 1 were selected with certainty (probability 1).

⁴ This study focuses on large jails, given that, as mentioned previously, they are the ones that were more likely being sampled in all the years of ASJ during the study period, which allows for a more consistent observation of the trends in racial disparities in jail incarceration. Large jails can also serve better in representing overall jail populations in the states of interest. Although some small county jails also likely appeared in all waves of ASJ being considered (the number of these jails is small), the race information contained in the ASJ data for these small jails tend to be highly skewed, likely adding bias to the results. In addition, it should be noted that jails in West Virginia are regional jails, each of which serves multiple counties. Given that these (10 regional jails in total) were all large jails with an ADP of over 300, they were included in the analysis, with incarceration rates for Black and White being calculated based on the total Black population and the total White population for the counties served by each of these regional jails.

⁵ The ASJ sample was drawn at the jurisdiction level. When a jail jurisdiction with multiple reporting units is sampled, data were collected from all reporting units within that jail jurisdiction.

⁶ In the SCM models examining Black-to-White disparities in jail incarceration between WA and the control states, the years of 2013, 2015 and 2019 were omitted. In the DiD analysis based on the combined data of CO and WA, we conducted two sets of analyses, one included all 13 years from 2007 to 2019, and one omitted the years of 2013, 2015 and 2019. The results from these two sets of analyses were substantively similar.

covariates (poverty level, median household income, unemployment rate, and percentage of men). θ_j is a fixed effect for each jail jurisdiction, and τ_t is a fixed effect for each year. As mentioned previously, the two-way fixed effects estimator helps control for time-invariant unobservable group characteristics. ε_{jt} is a jail jurisdiction-year-specific error term. Our unit of analysis for the DiD models is a jail jurisdiction-year observation.

It should be noted that the DiD approach relies upon the common trend assumption, which states that the average outcomes for the treatment and control groups, absent the treatment itself, would have followed the same time trends. Specific to the current study, the DiD approach requires a valid counterfactual to county jails in CO and WA to measure the effects of this drug policy change. However, the county jails considered in the study may vary considerably and in non-overlapping ways. Thus, a matching approach, namely the SCM was used to further explore the trends in Black–White jail incarceration disparity following legalization in large jails in the two legalized states. The SCM results help demonstrate how CO's and WA's Black–White jail incarceration disparities changed, as compared to “synthetic CO” and “synthetic WA,” respectively, serving as control groups consisting of weighted values from the large county jails in the nonlegalized states. As a quasi-experimental design developed by [Abadie, Diamond, and Hainmueller \(2010\)](#) the SCM constructs a control group as a weighted average of non-treatment units in the pre-treatment period. This approach “allows for estimation of a specification with interactive fixed effects which nests more traditional additive fixed effects models” ([Powell, 2018](#), p. 2), and “accounts for endogeneity bias due to omitted variables by accounting for the presence of time-varying unobservable confounders” ([Olper, Curzi, and Swinnen, 2018](#), p. 395). As such, the SCM is ideal compared to traditional panel models which only account for time-invariant unobservable confounders. Following the approach described by [Abadie et al. \(2010\)](#), the synthetic control regions were constructed which best reproduced the values of predictors for the outcome variable in the pre-treatment/pre-legalization period. Specifically, the predictors included the ratios of Black jail incarceration rates to their White equivalent for the years in the pre-treatment period (2007–2012). The “donor pool” comprises large county jails in the nonlegalized states.⁷ The effects of legalization were estimated as the difference in the outcome variable between the treated regions (CO and WA) and their synthetic control versions in the years after the passage of the recreational cannabis laws in these two legalized states.⁸

Specifically, a synthetic control can be represented by a $J \times 1$ vector of weights, $W = (w_2 \dots, w_{J+1})$. The SCM chooses the synthetic control, $W^* = (w_2^* \dots, w_{J+1}^*)$ that minimizes the difference in the pre-treatment/legalization outcomes and covariates between the treatment/legalized units (CO or WA) and the average of the comparison/nonlegalized units:

$$(X_1 - X_0W)'V(X_1 - X_0W) \quad (2)$$

where X_1 and X_0 represent the pre-treatment outcomes and covariates for the treated unit (CO or WA) and control units (county jails in the nonlegalized states) respectively, and V captures the relative importance

⁷ We also estimated models with additional predictors such as county poverty level, median household income, unemployment rate, and percentage of men, given the outcome variable may be influenced by other factors related to county characteristics. Including these additional predictors resulted in synthetic control groups that closely resembled those used in the paper.

⁸ The SCM model used in this study allows the examination of one single treated unit and compare that with the weighted values from the nontreated units in the donor pool. To more accurately reflect racial disparities in CO and WA, the researcher first added up total number of Black inmates and White inmates, respectively, in counties of each treatment state each year; then, the Black jail rate and White jail rate for these two states were calculated by dividing the total number of inmates of corresponding race by the Black population and the White population, respectively, of the related counties.

of these variables in predicting the outcome of interest (for a detailed introduction to the synthetic control method, see [Abadie et al., 2010](#)). Information on the weights of donor pool jail jurisdictions for the models related to CO and WA is presented in Appendix Tables A1–A2.

4. Results

[Table 2](#) presents the effects of legalization on Black–White jail incarceration disparity in CO and WA based on the DiD models. The DiD results demonstrate that, although CO and WA seemed to experience a decrease in the Black–White jail incarceration disparity, this decrease appears minimal and did not reach statistical significance (the 95% confidence intervals cross 0). For the effects of the covariates, although median household income was found to be a statistically significant predictor of the outcome variable, its effect size was very small.

To further examine the Black–White jail incarceration gap post-legalization in large jails in the two legalized states, we used the SCM approach. [Figs. 1A to 2B](#) display the trends for Black–White disparity in jail incarceration rates for CO and WA, respectively, with calculations performed using the `synth` command in Stata 17. Note here that the post-legalization years for WA omitted 2013, 2015, and 2019, given the high degree of missingness in the WA data for these years. As these graphs display, the synthetic control regions closely tracked the CO and WA trajectories of the Black–White disparities in jail rates during the pre-legalization periods. The two trend lines for the comparisons between each of the legalized states and their corresponding synthetic control versions began to diverge after legalization, with post-legalization trend lines for the legalized states being overall lower than those for their SCM versions. This suggests that Black–White disparities have experienced some decrease in large jails in these two legalized states following legalization, relative to those in the non-legalized states.

The graphs generally suggest some reduction in racial disparities, but the parameter estimates must be assessed to determine if the reduction effect is of any substantive degree. To address this, a series of placebo tests were run using the `synth_runner` package in Stata 17 developed by [Galiani and Quistorff \(2017\)](#) to determine statistical significance. Specifically, the inference was obtained by estimating the same model on each untreated unit (jail jurisdictions in the non-legalized states) to get a distribution of placebo effects. The p -value (standardized) for each post-treatment period is the probability of a placebo standardized effect that is larger than or equal to the main estimate for the treated units (jail jurisdiction in CO/WA).⁹ As [Galiani and Quistorff \(2017\)](#) noted, if the distribution of the placebo effects produces many effects as large as the main estimate, then the estimated effect was likely to occur by chance. [Table 3](#) presents the post-treatment results from the SCM, with estimates provided for each post-legalization year. As [Table 3](#) shows, while the coefficients were negative for all post-legalization years in the model of WA, none indicated a probability of an effect beyond random chance. For CO, the only year in which the difference between the synthetic and actual Black–White disparity in jail rates was unusually large was 2014 ($p < 0.05$), suggesting that had CO not legalized cannabis, the ratio of Black jail rate to that for White would have been 0.626 higher that year. [Figs. 1B and 2B](#) virtually present the results of the placebo (permutation) test. As these graphs demonstrate, when we reassign exposure to legalization to the jail jurisdictions in the control states, there is a good chance of obtaining a placebo effect as large as the one obtained for CO/WA (a substantial proportion of grey lines representing the placebo effects in other regions are lower than the black line representing the estimated effect in CO/WA), indicating a generally insignificant impact

⁹ The placebo effect can be quite large if those units are not well-matched in the pretreatment period, resulting in overly conservative p -values ([Galiani and Quistorff, 2017](#)). One way to address this is to use the placebo standardized effect, which is obtained by dividing all effects by the corresponding pretreatment match quality (reflected by the pretreatment RMSPE).

Table 1

Summary statistics.

	Observation period: 2007–2019 (pre-leg. period: 2007–12; post-leg. period: 2013–19)				Observation period: 2007–12, 2014, 2016–18 ^a (pre-leg. period: 2007–12; post-leg. period: 2014, 2016–18)			
	Colorado		Control States		Washington		Control States	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Entire period	4.122	0.650	5.734	4.099	5.148	0.703	5.810	4.101
Pre-legalization period	4.685	0.522	6.128	4.094	5.608	0.470	6.128	4.094
Post-legalization period	3.640	0.174	5.405	4.077	4.457	0.235	5.347	4.072

^a This observation period is shorter given the substantial missing values for WA for the years 2013, 2015, and 2019.

of legalization on the outcome variable. Said another way, none of the placebo tests demonstrate an effect that is unusually large in the post-legalization period (if a statistically significant effect were observed, the black line would be immediately differentiated from the grey lines). Overall, the SCM and related placebo tests produced results¹⁰ that were substantively similar to those obtained from the DiD analyses. That is, despite observed reductions in Black–White disparities in jail rates following legalization in the two legalized states, these reductions were generally not large enough to be considered statistically significant.

5. Discussion

This study is the first to employ a quasi-experimental design via the SCM and the DiD analyses to investigate the impact of adult-use recreational cannabis legalization on Black–White disparities in jail incarceration. The results revealed no statistically significant reduction. Insubstantial effects were consistent in the large county jails across the two legalized states—CO and WA. Although there seemed to be some degree of reduction in Black–White jail incarceration differences after legalization, this decrease was not large enough to be considered a meaningful divergence. Overall, unfortunately, our data provide little support for the notion that legalization can solve the long-standing racial disparities in jail incarceration.

The study's findings align with previous research generally indicating that legalizing cannabis has little effect on reducing racial disparities in criminal legal system contacts, such as arrests for cannabis possession (Firth et al., 2019; Willits et al., 2022) and jail bookings (Stanton et al., 2020). Despite the observed social benefits of legalization, including a substantial decrease in the number of people across racial and ethnic backgrounds arrested for cannabis-related offenses (Sheehan et al., 2021; Wu and Cullenbine, 2022), the associated avoidance of collateral consequences of a drug-related arrest or conviction, and potential therapeutic benefits (McNabb et al., 2023), legalization at the state level is not enough to eliminate existing racial disparities in legal system involvement. In effect, as our data indicated, despite the overall declining trends, the Black jail incarceration rates remain around 4 times greater than the White jail incarceration rates in CO and WA at the end of the observation period (2019 for CO and 2018 for WA). The observed small (insignificant) reduction in Black–White jail incarceration disparities in the two legalized states may not be interpreted as negligible, but they challenge the notion of a potentially large, meaningful decline in racial disparities in incarceration resulting from legalization, like drug law reform advocates (e.g., ACLU, and Law Enforcement Against Prohibition) have hoped.

The persisting racial disparities in jail bookings could be attributable to a myriad of factors including potential differences among racial groups in their engagement with the illegal cannabis market, bias in law enforcement (Beckett et al., 2005; Gaston, 2019), and the common

Table 2

DiD results on Black–White disparity in jail incarceration rates in Colorado and Washington State

Predictors	Estimate (SE)	95% Confidence interval
Legalization	−0.052 (0.320)	[−0.680, 0.577]
Covariates		
Poverty	0.038 (0.033)	[−0.027, 0.103]
Median income	0.00005 (0.0002)	[9.12e-06, 0.00008]
Unemployment	−0.028 (0.050)	[−0.125, 0.069]
Percentage of men	−0.050 (0.193)	[−0.429, 0.330]
Jail jurisdiction dummies	YES	–
Year dummies	YES	–
N	1952	–
R ²	0.82	–

Note. CO and WA legalized recreational marijuana in late 2012. Jail jurisdiction dummies and year dummies were included in the model to control for jail jurisdiction-year fixed effects. Given the high missingness, data for jail jurisdictions in WA does not contain 2013, 2015, and 2019.

cultural script that unjustly associates Blackness with drug use (Alexander, 2010; Beckett et al., 2005; Lindsay and Vuolo, 2021). As scholars have noted, illegal cannabis markets remain popular despite legalization (Bender, 2016; Stohr et al., 2020), and Black Americans are more likely to be arrested for cannabis distribution (selling) than Whites (Firth et al., 2019) despite similar rates of drug selling across different racial and ethnic groups (Jones, Logan, Gladden, and Bohm, 2015). Notably, the profitable new legal cannabis industry is not “a level playing field for anyone interested in the market; it is a situation where blacks are prevented from participating” (Danquah-Brobby, 2016, p. 534). Barriers for Black Americans to enter into the legal cannabis industry include criminal background checks for licensure applications, substantial monetary resources needed to be eligible for a license, and farming requirements that need to be met to qualify as a cultivator. Perhaps ironically, or through willful design, the historical criminalization of cannabis and disproportionate convictions for cannabis-related crimes may culminate into the very criminal history that prevents Black Americans from participating in this lucrative legal industry. In this way, the ostensibly colorblind War on Drugs and cannabis, in particular, continues to promote a Black underclass largely locked out of an industry that holds the potential to remedy some of the harm and damage that has been done to Black families and communities and continues to promote a massive racial wealth gap (Connley, 2021; Danquah-Brobby, 2016). In addition, lower-class Black Americans are likely to be more sensitive to cannabis price changes created by the legalization of this drug and may tend to buy this drug from the illicit cannabis market (as the price of illicit cannabis is lower than legal cannabis), resulting in a higher chance of being arrested. It should also be noted that the size of the overall jail population has not substantially declined in some of the states that have legalized recreational cannabis (Stanton et al., 2020), indicating that the police may shift their efforts toward increasing arrests for other offenses. This undermines the ability for cannabis legalization to meaningfully reduce mass incarceration and, by extension, any collateral consequences such as reduced housing access and employment opportunities that are associated with arrest and conviction for a criminal offense. Since legalization itself does not address racial

¹⁰ We conducted additional analyses to ensure the reliability of the SCM results. This involved re-analyzing the data without the donor county jails with the largest weight. The results of these additional analyses were largely consistent with those obtained from the original SCM models.

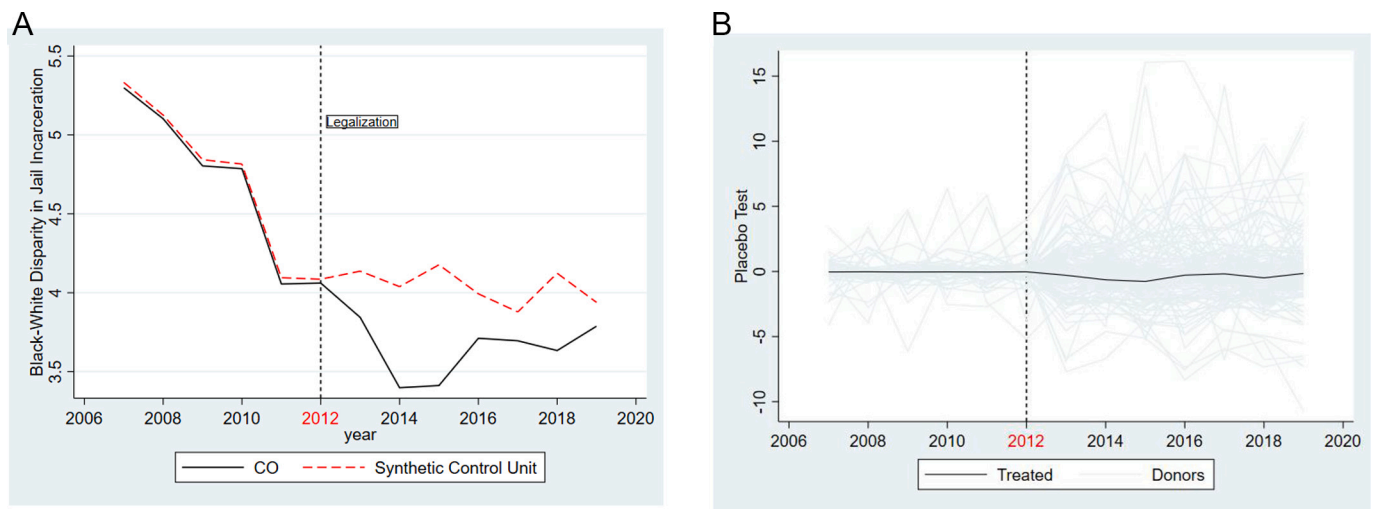


Fig. 1. (A) Synthetic control model of Black–White disparity in jail incarceration rates in Colorado. (B) Distribution of placebo effects (Black–White disparity in jail incarceration in Colorado).

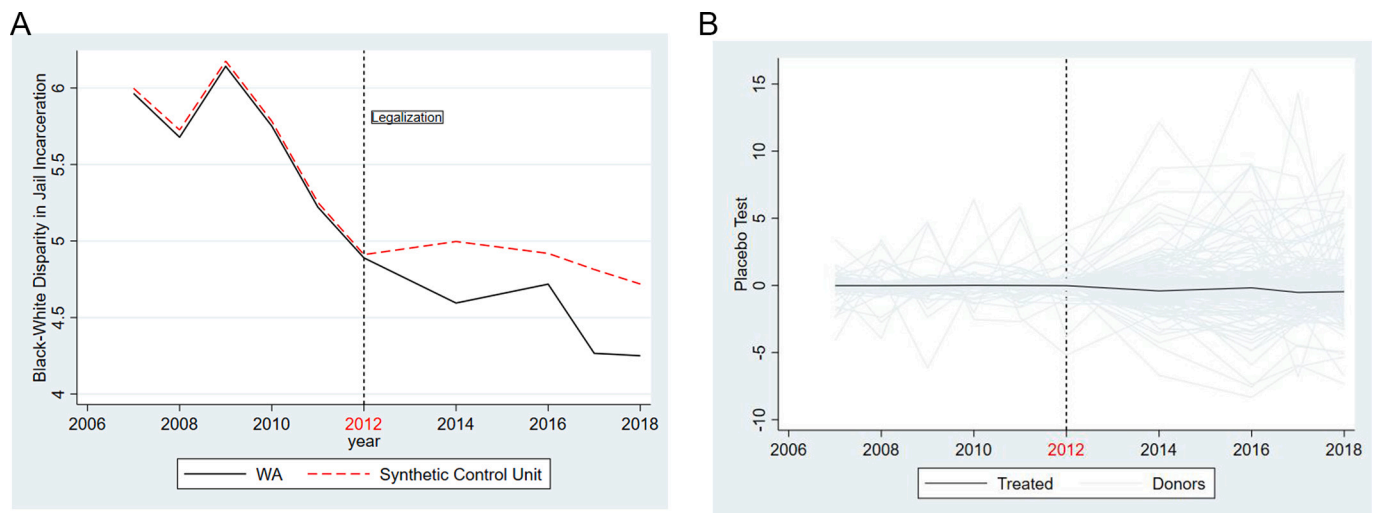


Fig. 2. (A) Synthetic Control Model of Black–White Disparity in Jail Incarceration Rates in Washington State. *Note.* The observation period for WA covers 2007–2012, 2014, and 2016–2018. The years 2013, 2015, and 2019 were omitted due to the high missingness. (B) Distribution of Placebo Effects (Black–White Disparity in Jail Incarceration in Washington State).

bias (implicit or explicit) in police behavior, including the use of arrest discretion, it is not surprising to see minimal changes in racial disparity in jail incarceration rates in the legalized states.

Racial disparity in incarceration has long been a significant concern. Scholars have offered multiple theoretical perspectives to understand this phenomenon, including conflict and social threat theories (Wortley and Tanner, 2005), the perspective of implicit bias within law enforcement (Weisbuch, Pauker, and Ambady, 2009), and differential involvement in crime hypothesis (Hindelang, 1978), though mixed and inconclusive results have been yielded from research on the differential involvement hypothesis (see Unnever, Owusu-Bempah, and Deryol, 2019). Indeed, racial disparity in jail incarceration, and more broadly in the criminal legal system, is a social problem with deep roots in the nation's economic and social structures that cannot be easily or quickly addressed by a single drug policy change like the legalization of recreational cannabis. As scholars have noted, these roots are embedded in America's "always-already racialized" political economy, or "a form of capitalism and democratic governance forged on the anvils of slavery and racial segregation and rendered increasingly insecure and exclusionary in the decades following deindustrialization" (Garland, 2023, p.

44), which has long distributed economic, political, social, and other resources to groups along racial lines (Bell, 2004; Golash-Boza, 2015). Thus, communities of color have varying experiences shaped by race/racism, including in their interactions with the criminal legal system (Peterson, Krivo, and Hagan, 2006). Indeed, a considerable body of research has documented that the penal system in America is largely a system of racialized social control that creates and maintains racial hierarchy or inequality by disproportionately arresting, prosecuting, and convicting/incarcerating people of racial/ethnic minority groups, especially Black Americans (Alexander, 2010; Berry, 1995; Davis, 2009; Muhammad, 2019; Tonry, 2010).¹¹ Given that racial disparities in

¹¹ Many police practices were found to direct disproportionately toward Black Americans and other people of color, such as stop-and-frisk, police deployments, arrest decisions, and shooting decisions (for a review, see Roussell et al., 2019). In addition, sentencing policies such as mandatory minimums (especially the notorious 100:1 sentencing disparity between crack and power cocaine) associated with "War on Drugs" disproportionately affected disadvantaged minority communities (for a review, see Mauer, 2011).

Table 3

SCM results on Black–White disparity in jail incarceration rates in Colorado and Washington State

Post-legalization years	Colorado estimates	Washington estimates
2013	−0.293	–
2014	−0.640*	−0.410
2015	−0.765	–
2016	−0.281	−0.180
2017	−0.183	−0.523
2018	−0.490	−0.470
2019	−0.150	–

Note. * $p < .05$.

incarceration and other criminal legal system contacts are essentially a function of larger structural arrangements (with racism and inequality embedded in the political economy) in American society, substantial progress toward racial equality within the criminal legal system can only be made when these arrangements are altered. Certainly, this is not easy. To make a real difference, there will be a need for considerable efforts to promote social consensus about the role of race/racism in defining the basic structure of American society and to address the structural determinants of socioeconomic disadvantages attributable to people of color (Garland, 2023; Willits et al., 2022).

5.1. Limitations

This study provides critical insight into the effects of cannabis legalization on racial disparities in jail incarceration. However, it is not without limitations. First, although we focused on CO and WA, the first pair of states that legalized recreational cannabis in the county, which allows for an extended post-legalization period to examine the impact of legalization on Black–White jail incarceration disparities, our post-legalization period covered only 7 years (with the observation period for WA further shortened in some analyses because of missing data). Hence, the effects identified in this study are largely short-term effects. Given that the magnitude of the impact of legalization on criminal legal system outcomes may change over time (Hall and Lynskey, 2016), future research should further examine the influence of legalization on racial disparities in imprisonment when additional years of data are available. Secondly, this study prioritizes large jails as they offer a more representative sample of the broader jail populations, particularly in terms of the skewed racial data inherent in the ASJ dataset for smaller jails. However, we acknowledge that a comprehensive analysis involving jails of varying sizes would provide a more nuanced understanding of the impact of legalization on racial disparities in jail incarceration. To overcome the constraints of the ASJ data, researchers seeking to delve deeper into this matter may find it necessary to directly gather jail

population data from all categories of county jails in legalized states to model jail incarceration trends. Third, while this study controlled for four socioeconomic characteristics in our multivariate analyses, we acknowledge that our models did not incorporate other theoretically relevant factors such as violent crime rates, law enforcement officer rates, and racial composition. We recommend that future research broaden their models to include these additional factors and account for their effects on the outcome variable. Fourth, while this study used a quasi-experimental design with a combination of DiD and SCM that helps identify the causal relationship, it is not a true experiment. Therefore, we caution against overinterpretation of the results. Given the very limited literature on this subject, more research based on rigorous impact assessment methodologies is needed to accumulate evidence unveiling the implications of recreational cannabis legalization for racial disparities in incarceration.

6. Conclusion

In conclusion, this study explored the critical issue of the impact of adult-use recreational cannabis legalization on racial disparities in the jail population. While some have hoped that the legalization of cannabis would provide an effective means of reducing jail population disparities, these findings do not support that conclusion. This study provides valuable evidence and provides a start, but research must continue to explore the impact of the legalization of cannabis on racial disparities in the criminal legal system. The lack of significant findings perhaps indicates more the complexities in criminal justice contact and deep-rooted nature of racial disparity in the criminal legal system rather than a weakness in legalization. Ultimately, multi-faceted strategies and solutions beyond legalization that address racialized social control and racism and inequality in America are called for. Until then, the struggle to eradicate racial disparities in the criminal legal system will continue.

CRediT authorship contribution statement

Guangzhen Wu: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Katherine A. Durante:** Writing – review & editing, Writing – original draft, Investigation. **Heather C. Melton:** Writing – review & editing, Writing – original draft, Investigation.

Declaration of competing interest

This work was supported by the Office of Vice President for Research and the College of Social and Behavioral Science of The University of Utah (Project No. 51900532).

Appendix A. Appendix

Table A1

Synthetic colorado jail jurisdiction weights for Black–White jail incarceration disparity

Donor jail jurisdictions	Weight	Donor jail jurisdictions	Weight	Donor jail jurisdictions	Weight	Donor jail jurisdictions	Weight
Montgomery (AL)	0.006	Richmond (GA)	0.006	Mercer (NJ)	0.002	Pennington (SD)	0.007
Russell (AL)	0.009	Spalding (GA)	0.004	Morris (NJ)	0.003	Blount (TN)	0.005
Talledega (AL)	0.005	Sumter (GA)	0.008	Ocean (NJ)	0.003	Greene (TN)	0.005
Coconino (AZ)	0.003	Thomas (GA)	0.004	Passaic (NJ)	0.003	Hamilton (TN)	0.006
Mohave (AZ)	0.003	Troup (GA)	0.004	Dona Ana (NM)	0.007	Knox (TN)	0.005
Pinal (AZ)	0.005	Canyon (ID)	0.002	Sandoval (NM)	0.002	Montgomery (TN)	0.004
Yuma (AZ)	0.006	Du Page (IL)	0.004	Albany (NY)	0.003	Putnam (TN)	0.003
Sebastian (AR)	0.006	Mchenry (IL)	0.002	Monroe (NY)	0.004	Robertson (TN)	0.003

(continued on next page)

Table A1 (continued)

Donor jail jurisdictions	Weight	Donor jail jurisdictions	Weight	Donor jail jurisdictions	Weight	Donor jail jurisdictions	Weight
Washington (AR)	0.005	Allen (IN)	0.004	Niagara (NY)	0.004	Shelby (TN)	0.005
Alachua (FL)	0.005	Elkhart (IN)	0.005	Orange (NY)	0.003	Sullivan (TN)	0.003
Broward (FL)	0.006	Vanderburgh (IN)	0.006	New York (NY)	0.003	Bell (TX)	0.004
Charlotte (FL)	0.005	Johnson (KS)	0.004	Cumberland (NC)	0.005	Collin (TX)	0.004
Lee (FL)	0.005	Sedgwick (KS)	0.004	Durham (NC)	0.004	Denton (TX)	0.006
Leon (FL)	0.003	Christian (KY)	0.006	Forsyth (NC)	0.003	Galveston (TX)	0.004
Manatee	0.003	Lexington-Fayette (KY)	0.007	Wake (NC)	0.003	Jefferson (TX)	0.004
Marion (FL)	0.005	Hardin (KY)	0.008	Greene (OH)	0.008	Johnson (TX)	0.006
Martin (FL)	0.003	Henderson (KY)	0.004	Lake (OH)	0.003	Liberty (TX)	0.007
Okaloosa (FL)	0.006	Kenton (KY)	0.006	Trumbull (OH)	0.003	Montgomery (TX)	0.005
Orange (FL)	0.006	Caddo Parish (LA)	0.006	Allegheny (PA)	0.004	Potter (TX)	0.005
Pasco (FL)	0.008	Calcasieu Parish (LA)	0.004	Berks (PA)	0.003	Travis (TX)	0.004
Pinellas (FL)	0.005	Catahoula Parish (LA)	0.005	Blair (PA)	0.003	Val Verde (TX)	0.022
Putnam (FL)	0.014	East Baton Rouge (LA)	0.008	Butler (PA)	0.002	Victoria (TX)	0.009
St Lucie (FL)	0.006	Morehouse Parish (LA)	0.005	Chester (PA)	0.004	Davis (UT)	0.003
Volusia (FL)	0.005	New Orleans Parish (LA)	0.003	Erie (PA)	0.004	Salt Lake (UT)	0.018
Bartow (GA)	0.152	Ouachita Parish (LA)	0.006	Franklin	0.003	Arlington (VA)	0.002
Bulloch (GA)	0.004	Terrebonne Parish Consolidated (LA)	0.004	Lackawanna (PA)	0.002	Fairfax (VA)	0.006
Chatham (GA)	0.004	Anne Arundel (MD)	0.004	Lancaster (PA)	0.003	Rockingham (VA)	0.006
Cherokee (GA)	0.002	Baltimore (MD)	0.006	Lebanon (PA)	0.002	Tygart Valley Regional Jail (WV)	0.002
Clarke (GA)	0.003	Prince Georges (MD)	0.004	Monroe (PA)	0.005	Eastern Regional Jail (WV)	0.005
Clayton (GA)	0.079	St Mary's (MD)	0.004	Montgomery (PA)	0.003	Western Regional Jail (WV)	0.007
Cobb (GA)	0.006	Washington (MD)	0.009	Northampton (PA)	0.004	Southwestern Regional Jail (WV)	0.005
Decatur (GA)	0.005	Baltimore (MD)	0.004	Philadelphia (PA)	0.004	Northern Regional Jail (WV)	0.004
Dougherty (GA)	0.006	Anoka (MN)	0.002	Pike (PA)	0.002	Southern Regional Jail (WV)	0.006
Effingham (GA)	0.003	Ramsey (MN)	0.004	Washington (PA)	0.004	Dane (WI)	0.002
Atlanta (GA)	0.043	Leake (MS)	0.008	Westmoreland (PA)	0.003	Kenosha (WI)	0.004
Glynn (GA)	0.004	St. Louis (MO)	0.004	York (PA)	0.003	Milwaukee (WI)	0.005
Henry (GA)	0.012	Douglas (NE)	0.003	Richland (SC)	0.003	Rock (WI)	0.003
Jefferson (GA)	0.006	Camden (NJ)	0.004	Spartanburg (SC)	0.004	Waukesha (WI)	0.002
Muscogee (GA)	0.006	Hudson (NJ)	0.002	Minnehaha (SD)	0.005	Winnebago (WI)	0.004

Notes. Root mean squared prediction error (RMSPE) = 2.06e-13. Calculations were performed using the *synth* command in Stata 17.

Table A2

Synthetic Washington jail jurisdiction weights for Black–White jail incarceration disparity

Donor jail jurisdictions	Weight	Donor jail jurisdictions	Weight	Donor jail jurisdictions	Weight	Donor jail jurisdictions	Weight
Montgomery (AL)	0.006	Richmond (GA)	0.006	Mercer (NJ)	0.004	Pennington (SD)	0.01
Russell (AL)	0.006	Spalding (GA)	0.006	Morris (NJ)	0.004	Blount (TN)	0.005
Talledega (AL)	0.006	Sumter (GA)	0.006	Ocean (NJ)	0.005	Greene (TN)	0.006
Coconino (AZ)	0.005	Thomas (GA)	0.007	Passaic (NJ)	0.005	Hamilton (TN)	0.006
Mohave (AZ)	0.005	Troup (GA)	0.006	Dona Ana (NM)	0.063	Knox (TN)	0.006
Pinal (AZ)	0.005	Canyon (ID)	0.006	Sandoval (NM)	0.006	Montgomery (TN)	0.007
Yuma (AZ)	0.005	Du Page (IL)	0.005	Albany (NY)	0.005	Putnam (TN)	0.005
Sebastian (AR)	0.006	Mchenry (IL)	0.004	Monroe (NY)	0.005	Robertson (TN)	0.005
Washington (AR)	0.005	Allen (IN)	0.005	Niagara (NY)	0.006	Shelby (TN)	0.004
Alachua (FL)	0.007	Elkhart (IN)	0.004	Orange (NY)	0.005	Sullivan (TN)	0.006
Broward (FL)	0.006	Vanderburgh (IN)	0.005	New York (NY)	0.005	Bell (TX)	0.006
Charlotte (FL)	0.005	Johnson (KS)	0.005	Cumberland (NC)	0.006	Collin (TX)	0.006
Lee (FL)	0.006	Sedgwick (KS)	0.005	Durham (NC)	0.006	Denton (TX)	0.008
Leon (FL)	0.004	Christian (KY)	0.006	Forsyth (NC)	0.005	Galveston (TX)	0.005
Manatee	0.006	Lexington-Fayette (KY)	0.008	Wake (NC)	0.005	Jefferson (TX)	0.006
Marion (FL)	0.006	Hardin (KY)	0.006	Greene (OH)	0.006	Johnson (TX)	0.006
Martin (FL)	0.005	Henderson (KY)	0.006	Lake (OH)	0.005	Liberty (TX)	0.006
Okaloosa (FL)	0.006	Kenton (KY)	0.005	Trumbull (OH)	0.005	Montgomery (TX)	0.007
Orange (FL)	0.006	Caddo Parish (LA)	0.006	Allegheny (PA)	0.005	Potter (TX)	0.006
Pasco (FL)	0.006	Calcasieu Parish (LA)	0.006	Berks (PA)	0.005	Travis (TX)	0.006
Pinellas (FL)	0.006	Catahoula Parish (LA)	0.006	Blair (PA)	0.004	Val Verde (TX)	0.004
Putnam (FL)	0.006	East Baton Rouge (LA)	0.006	Butler (PA)	0.003	Victoria (TX)	0.005
St Lucie (FL)	0.006	Morehouse Parish (LA)	0.021	Chester (PA)	0.004	Davis (UT)	0.004
Volusia (FL)	0.006	New Orleans Parish (LA)	0.006	Erie (PA)	0.005	Salt Lake (UT)	0.015
Bartow (GA)	0.016	Ouachita Parish (LA)	0.006	Franklin	0.004	Arlington (VA)	0.004
Bulloch (GA)	0.026	Terrebonne Parish Consolidated (LA)	0.005	Lackawanna (PA)	0.004	Fairfax (VA)	0.005
Chatham (GA)	0.01	Anne Arundel (MD)	0.006	Lancaster (PA)	0.004	Rockingham (VA)	0.006
Cherokee (GA)	0.004	Baltimore (MD)	0.006	Lebanon (PA)	0.004	Tygart Valley Regional Jail (WV)	0.004
Clarke (GA)	0.005	Prince Georges (MD)	0.006	Monroe (PA)	0.006	Eastern Regional Jail (WV)	0.005
Clayton (GA)	0.01	St Mary's (MD)	0.006	Montgomery (PA)	0.005	Western Regional Jail (WV)	0.006
Cobb (GA)	0.006	Washington (MD)	0.007	Northampton (PA)	0.006	Southwestern Regional Jail (WV)	0.006
Decatur (GA)	0.007	Baltimore (MD)	0.005	Philadelphia (PA)	0.005	Northern Regional Jail (WV)	0.004
Dougherty (GA)	0.006	Anoka (MN)	0.004	Pike (PA)	0.004	Southern Regional Jail (WV)	0.013
Effingham (GA)	0.007	Ramsey (MN)	0.005	Washington (PA)	0.006	Dane (WI)	0.004
Atlanta (GA)	0.005	Leake (MS)	0.007	Westmoreland (PA)	0.004	Kenosha (WI)	0.005
Glynn (GA)	0.006	St. Louis (MO)	0.005	York (PA)	0.005	Milwaukee (WI)	0.005
Henry (GA)	0.005	Douglas (NE)	0.004	Richland (SC)	0.005	Rock (WI)	0.004

(continued on next page)

Table A2 (continued)

Donor jail jurisdictions	Weight	Donor jail jurisdictions	Weight	Donor jail jurisdictions	Weight	Donor jail jurisdictions	Weight
Jefferson (GA)	0.008	Camden (NJ)	0.006	Spartanburg (SC)	0.005	Waukesha (WI)	0.003
Muscogee (GA)	0.006	Hudson (NJ)	0.009	Minnehaha (SD)	0.031	Winnebago (WI)	0.004

Notes. Root mean squared prediction error (RMSPE) = 3.47e-13. Calculations were performed using the *synth* command in Stata 17.

References

- Abadie, A., Diamond, A., & Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. *Journal of the American Statistical Association*, 105(490), 493–505. <https://doi.org/10.1198/jasa.2009.ap08746>
- Alexander, M. (2010). *The New Jim Crow: Mass Incarceration in the Age of Colorblindness*. The New Press.
- American Civil Liberties Union. (2020). *A tale of two countries: Racially targeted arrests in the era of marijuana reform (research report)*. <https://www.aclu.org/report/tale-two-countries-racially-targeted-arrests-era-marijuana-reform>.
- Beckett, K., Nyrop, K., Pfingst, L., & Bowen, M. (2005). Drug use, drug possession arrests, and the question of race: Lessons from Seattle. *Social Problems*, 52(3), 419–441. <https://doi.org/10.1525/sp.2005.52.3.419>
- Bell, D. (2004). *Silent Covenants: Brown v. Board of Education and the Unfulfilled Hopes for Racial Reform*. Oxford University Press.
- Bender, S. W. (2016). The colors of cannabis: Race and marijuana. *U.C. Davis Law Review*, 50, 689.
- Berry, M. F. (1995). *Black Resistance/White law: A History of Constitutional Racism in America*. New York: Penguin Press.
- Brenan, M. (2023, November 16). *Americans more critical of U.S. criminal justice system*. *Gallup*. <https://news.gallup.com/poll/544439/americans-critical-criminal-justice-system.aspx#:~:text=Story%20Highlights&text=WASHINGTON%2C%20D.C.%20%2D%20A%2058,low%2041%25%20said%20the%20same..>
- Brinkman, J., & Mok-Lamme, D. (2019). Not in my backyard? Not so fast. The effect of marijuana legalization on neighborhood crime. *Regional Science and Urban Economics*, 78, Article 103460. <https://doi.org/10.1016/j.regsciurbeco.2019.103460>
- Connley, C. (2021). *Cannabis is projected to be a \$70 billion market by 2028—Yet those hurt most by the war on drugs lack access*. *CNBC*. <https://www.cnn.com/2021/07/01/i-n-billion-dollar-cannabis-market-racial-inequity-persists-despite-legalization.html>.
- Contreras, C. (2017). A block-level analysis of medical marijuana dispensaries and crime in the city of Los Angeles. *Justice Quarterly*, 34(6), 1069–1095. <https://doi.org/10.1080/07418825.2016.1270346>
- Cooper, W. L., & Thompson, C. (2020, November 11). *Will drug legalization leave Black people behind? The Marshall Project*. <https://www.themarshallproject.org/2020/11/11/will-drug-legalization-leave-black-people-behind>.
- Danquah-Brobby, E. (2016). Prison for you: Profit for me: Systemic racism effectively bars blacks from participation in newly-legal marijuana industry. *University of Baltimore Law Review*, 46, 523.
- Davis, A. J. (2009). *Arbitrary Justice: The Power of the American Prosecutor*. Oxford University Press.
- Dills, A., Goffard, S., Miron, J., & Partin, E. (2021). *The effect of state marijuana legalizations: 2021 update*. *Cato Institute*. <https://www.cato.org/policy-analysis/effect-state-marijuana-legalizations-2021-update>.
- Dragone, D., Prarolo, G., Vanin, P., & Zanella, G. (2019). Crime and the legalization of recreational marijuana. *Journal of Economic Behavior & Organization*, 159, 488–501. <https://doi.org/10.1016/j.jebo.2018.02.005>
- Durante, K. A. (2020). Racial and ethnic disparities in prison admissions across counties: An evaluation of racial/ethnic threat, socioeconomic inequality, and political climate explanations. *Race and Justice*, 10(2), 176–202. <https://doi.org/10.1177/2153368717738038>
- Durante, K. A., Trocchio, S., & Martínez, C. (2024). Doomed by the data? Latina/o/x racialization and sentencing outcomes in Florida. *Critical Criminology*. <https://doi.org/10.1007/s10612-024-09766-4>. Advance online publication.
- Ellison, J. M., & Spohn, R. E. (2017). Borders up in smoke: Marijuana enforcement in Nebraska after Colorado's legalization of medicinal marijuana. *Criminal Justice Policy Review*, 28(9), 847–865. <https://doi.org/10.1177/0887403415615649>
- Farley, E. J., & Orchowsky, S. (2019). *Measuring the Criminal Justice System Impacts of Marijuana Legalization and Decriminalization Using State Data*. JRSA, Justice Research and Statistics Association.
- Firth, C. L., Maher, J. E., Dilley, J. A., Darnell, A., & Lovrich, N. P. (2019). Did marijuana legalization in Washington State reduce racial disparities in adult marijuana arrests? *Substance Use & Misuse*, 54(9), 1582–1587. <https://doi.org/10.1080/10826084.2019.1593007>
- Freisthler, B., Gaidus, A., Tam, C., Ponicki, W. R., & Gruenewald, P. J. (2017). From medical to recreational marijuana sales: Marijuana outlets and crime in an era of changing marijuana legislation. *The Journal of Primary Prevention*, 38, 249–263. <https://doi.org/10.1007/s10935-017-0472-9>
- Galiani, S., & Quistorff, B. (2017). The *synth_runner* package: Utilities to automate synthetic control estimation using *synth*. *The Stata Journal*, 17(4), 834–849.
- Garland, D. (2023). The current crisis of American criminal justice: A structural analysis. *Annual Review of Criminology*, 6, 43–63. <https://doi.org/10.1146/annurev-criminol-030722-035139>
- Gaston, S. (2019). Producing race disparities: A study of drug arrests across place and race. *Criminology*, 57(3), 424–451. <https://doi.org/10.1111/1745-9125.12207>
- Golash-Boza, T. (2015). Targeting Latino men: Mass deportation from the USA, 1998–2012. *Ethnic and Racial Studies*, 38(8), 1221–1228. <https://doi.org/10.1080/01419870.2015.988739>
- Green, T. V. (2022). *Americans overwhelmingly say marijuana should be legal for medical or recreational use*. *Pew Research Center*. <https://www.pewresearch.org/short-reads/2022/11/22/americans-overwhelmingly-say-marijuana-should-be-legal-for-medical-or-recreational-use/>.
- Hall, W., & Lynskey, M. (2016). Evaluating the public health impacts of legalizing recreational cannabis use in the United States. *Addiction*, 111(10), 1764–1773. <https://doi.org/10.1111/add.13428>
- Hao, Z., & Cowan, B. W. (2020). The cross-border spillover effects of recreational marijuana legalization. *Economic Inquiry*, 58(2), 642–666. <https://doi.org/10.1111/ecin.12764>
- Harper, A. J., & Jorgensen, C. (2023). Crime in a time of cannabis: Estimating the effect of legalizing marijuana on crime rates in Colorado and Washington using the synthetic control method. *Journal of Drug Issues*, 53(4), 552–580. <https://doi.org/10.1177/00220426221134107>
- Hindelang, M. J. (1978). Race and involvement in common law personal crimes. *American Sociological Review*, 43, 93–109.
- Jones, C. M., Logan, J., Gladden, R. M., & Bohm, M. K. (2015). Vital signs: Demographic and substance use trends among heroin users—United States, 2002–2013. *Morbidity and Mortality Weekly Report*, 64(26), 719–725.
- Jorgensen, C., & Harper, A. J. (2020). Examining the effects of legalizing marijuana in Colorado and Washington on clearance rates: A quasi-experimental design. *Journal of Experimental Criminology*, 1–22. <https://doi.org/10.1007/s11292-020-09446-7>
- Joshi, S., Doonan, S. M., & Pamplin, J. R., II (2023). A tale of two cities: Racialized arrests following decriminalization and recreational legalization of cannabis. *Drug and Alcohol Dependence*, 1, Article 109911. <https://doi.org/10.1016/j.drugalcdep.2023.109911>
- Kreit, A. (2016). Marijuana legalization and pretextual stops. *U.C. Davis Law Review*, 50, 741–772.
- Lantz, B., Wenger, M. R., & Craig, C. J. (2023). What if they were White? The differential arrest consequences of victim characteristics for Black and White co-offenders. *Social Problems*, 70(2), 297–320. <https://doi.org/10.1093/socpro/spab043>
- Lieb, D. A. (2024, May 4). *As the US Moves to Reclassify Marijuana as a Less Dangerous Drug, Could More States Legalize It?* Associated Press. <https://apnews.com/article/marijuana-reclassification-recreational-medical-states-83b1ad0e01bcd65142ca6cf4abdd110b>
- Lindsay, S. L., & Vuolo, M. (2021). Criminalized or medicalized? Examining the role of race in responses to drug use. *Social Problems*, 68(4), 942–963. <https://doi.org/10.1093/socpro/spab027>
- LoBianco, T. (2016, March 24). *Report: Aide Says Nixon's War on Drugs Targeted Blacks, Hippies*. *CNN*. <https://www.cnn.com/2016/03/23/politics/john-ehrllichman-richard-nixon-drug-war-blacks-hippie/index.html>
- Lu, R., Willits, D., Stohr, M. K., Makin, D., Snyder, J., Lovrich, N., ... Hemmens, C. (2021). The cannabis effect on crime: Time-series analysis of crime in Colorado and Washington State. *Justice Quarterly*, 38(4), 565–595. <https://doi.org/10.1080/07418825.2019.1666903>
- Maier, S. L., Mannes, S., & Koppenhofer, E. L. (2017). The implications of marijuana decriminalization and legalization on crime in the United States. *Contemporary Drug Problems*, 44(2), 125–146. <https://doi.org/10.1177/0091450917708790>
- Makin, D. A., Willits, D. W., Wu, G., DuBois, K. O., Lu, R., Stohr, M. K., ... Lovrich, N. P. (2019). Marijuana legalization and crime clearance rates: Testing proponent assertions in Colorado and Washington State. *Police Quarterly*, 22(1), 31–55. <https://doi.org/10.1177/109861118786255>
- Mauer, M. (2011). Addressing racial disparities in incarceration. *The Prison Journal*, 91(3), 87S–101S.
- McNabb, M., Durante, K. A., Trocchio, S., Ritter, D. J., MacCaffrie, R., Brum, A., ... White, S. (2023). Self-reported medicinal cannabis use as an alternative to prescription and over-the-counter medication use among US military veterans. *Clinical Therapeutics*, 45(6), 562–577. <https://doi.org/10.1016/j.clinthera.2023.04.003>
- Meize, M. R., Stohr, M. K., Willits, D. W., Solensten, B., Hampton, M. M., Makin, D. A., & Stanton, D. L., Sr. (2022). The intersection of gender, race, and arrest in the era of cannabis legalization. *Crime & Delinquency*, 70(6–7), 1639–1662. <https://doi.org/10.1177/00111287221083892>
- Muhammad, K. G. (2019). *The Condemnation of Blackness: Race, Crime, and the Making of Modern Urban America, with a New Preface*. Cambridge, MA: Harvard University Press.
- Olper, A., Curzi, D., & Swinnen, J. (2018). Trade liberalization and child mortality: A synthetic control method. *World Development*, 110, 394–410. <https://doi.org/10.1016/j.worlddev.2018.05.034>
- Peterson, R. D., Krivo, L. J., & Hagan, J. (2006). *The Many Colors of Crime: Inequalities of Race, Ethnicity, and Crime in America*. New York University Press.
- Pierson, E., Simoiu, C., Overgoor, J., Corbett-Davies, S., Jensen, D., Shoemaker, A., & Goel, S. (2020). A large-scale analysis of racial disparities in police stops across the

- United States. *Nature Human Behaviour*, 4(7), 736–745. <https://doi.org/10.1038/s41562-020-0858-1>
- Powell, D. (2018). *Imperfect Synthetic Controls: Did the Massachusetts Health Care Reform Save Lives?* WR-1246. RAND Corporation.
- Roussell, A., Henne, K., Glover, K. S., & Willits, D. (2019). Impossibility of a reverse racism effect: A rejoinder to James, James, and Vila. *Criminology & Public Policy*, 18, E5–E16. <https://doi.org/10.1111/1745-9133.12289>
- Sawyer, W., & Wagner, P. (2023). *Mass incarceration: The whole pie 2023* Prison Policy Initiative. <https://www.prisonpolicy.org/reports/pie2023.html>.
- Sheehan, B. E., Gruzca, R. A., & Plunk, A. D. (2021). Association of racial disparity of cannabis possession arrests among adults and youths with statewide cannabis decriminalization and legalization. *JAMA Health Forum*, 2(10), Article e213435. <https://doi.org/10.1001/jamahealthforum.2021.3435>
- Stanton, D., Mei, X., Kim, S., Willits, D., Stohr, M., Hemmens, C., ... Lovrich, N. (2020). The effect of marijuana legalization on jail populations in Washington State. *The Prison Journal*, 100(4), 510–530. <https://doi.org/10.1177/0032885520939313>
- Stohr, M., Makin, D., Stanton, D., Hemmens, C., Willits, D., Lovrich, N., & Wu, G. (2020). An evolution rather than a revolution: Cannabis legalization implementation from the perspective of the police in Washington State. *Justice Evaluation Journal*, 3(2), 267–293. <https://doi.org/10.1080/24751979.2020.1756378>
- Tonry, M. (1995). *Malign Neglect: Race, Crime, and Punishment in America*. Oxford University Press.
- Tonry, M. (2010). The social, psychological, and political causes of racial disparities in the American criminal justice system. *Crime and Justice*, 39(1), 273–312. <https://doi.org/10.1086/653045>
- Tran, N. K., Goldstein, N. D., Purtle, J., Massey, P. M., Lankenau, S. E., Suder, J. S., & Tabb, L. P. (2020). The heterogeneous effect of marijuana decriminalization policy on arrest rates in Philadelphia, Pennsylvania, 2009–2018. *Drug and Alcohol Dependence*, 212, Article 108058. <https://doi.org/10.1016/j.drugalcdep.2020.108058>
- Unnever, J., Owusu-Bempah, A., & Deryol, R. (2019). A test of the differential involvement hypothesis. *Race and Justice*, 9(2), 197–224.
- Weisbuch, M., Pauker, K., & Ambady, N. (2009). The subtle transmission of race bias via televised nonverbal behavior. *Science*, 326, 1711–1714.
- Western, B. (2006). *Punishment and Inequality in America*. Russell Sage Foundation.
- Willits, D. W., Solensten, B., Meize, M., Stohr, M. K., Makin, D. A., Hemmens, C., & Lovrich, N. P. (2022). Racial disparities in the wake of cannabis legalization: documenting persistence and change. *Race and Justice*. <https://doi.org/10.1177/21533687221087355>. Advance online publication.
- Wortley, S., & Tanner, J. (2005). Inflammatory rhetoric? Baseless accusations? Responding to Gabor's critique of racial profiling research in Canada. *Canadian Journal of Criminology*, 47, 581–609.
- Wu, G., Boateng, F. D., & Lang, X. (2020). The spillover effect of recreational marijuana legalization on crime: Evidence from neighboring states of Colorado and Washington State. *Journal of Drug Issues*, 50(4), 392–409. <https://doi.org/10.1177/0022042620921359>
- Wu, G., & Cullenbine, R. R. (2022). Recreational marijuana legalization and drug-related offenses in Washington State: An interrupted time series analysis with a combination of synthetic controls. *Journal of Experimental Criminology*, 20, 395–420. <https://doi.org/10.1007/s11292-022-09539-5>
- Wu, G., Durante, K. A., Melton, H. C., & Roddy, A. L. (2023). Marijuana legalization and racial disparities in prison incarceration in Colorado and Washington State: A quasi-experimental study. *Journal of Experimental Criminology*. Advance online publication. <https://doi.org/10.1007/s11292-023-09590-w>
- Wu, G., Li, Y., & Lang, X. (2022). Effects of recreational marijuana legalization on clearance rates for violent crimes: Evidence from Oregon. *International Journal of Drug Policy*, 100, Article 103528. <https://doi.org/10.1016/j.drugpo.2021.103528>
- Wu, G., Wen, M., & Wilson, F. A. (2021). Impact of recreational marijuana legalization on crime: Evidence from Oregon. *Journal of Criminal Justice*, 72, Article 101742. <https://doi.org/10.1016/j.jcrimjus.2020.101742>
- Wu, G., & Willits, D. W. (2022). The impact of recreational marijuana legalization on simple assault in Oregon. *Journal of Interpersonal Violence*, 37(23–24), NP23180–NP23201. <https://doi.org/10.1177/08862605221076169>
- Zellers, S. M., Ross, J. M., Saunders, G. R., Ellingson, J. M., Walvig, T., Anderson, J. E., & Vrieze, S. (2023). Recreational cannabis legalization has had limited effects on a wide range of adult psychiatric and psychosocial outcomes. *Psychological Medicine*, 53(14), 1–10. <https://doi.org/10.1017/S0033291722003762>